

FLEXHAND · PROBLEM STATEMENT

FlexHand problem statement

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	FXH-PRB-001	Version	0.5
Status	Draft	Date	2026-10-01
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/flexhand		
Source	docs/01-problem.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-01	Amish Chadha	Parts cost stated as a value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); first severity band set to mild flexor tone
0.3	2026-09-25	Amish Chadha	Record decisions of FXH-DDR-001 (passive motion only, design load); complete the passive-mobilization citation; severity band finding from FXH-CAL-001
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (problem, users, context, constraints, prior work)
0.1	2026-09-24	Amish Chadha	Initial scaffold

FlexHand problem statement

Recovery of hand function after stroke seems to depend on a large number of movement repetitions, but a typical therapy session delivers only a few dozen, and the rest of the day the affected hand often stays still and curled into flexion. FlexHand is a low-cost, open, wearable device that moves the fingers through hundreds of controlled flexion and extension cycles a day at home, within limits a therapist sets.

■ The problem

Hand impairment is common and persistent after stroke. In a cohort of people who had a flaccid arm soon after stroke, only about 38 % had regained some dexterity at six months ([Kwakkel et al., *Stroke*, 2003](#)). Difficulty opening the hand is a large part of this impairment: finger extension deficits after stroke arise from both neural factors and changed muscle mechanics ([Kamper and Rymer, *Muscle & Nerve*, 2003](#)), and finger extension is typically affected more than flexion ([Soft pneumatic actuators for pushing fingers into extension, *JNER*, 2024](#)).

Therapy dose is small. Lang and colleagues observed 312 physical and occupational therapy sessions and found an average of 32 repetitions of upper-limb functional movement per session, against the 400 to 600 repetitions per session used in animal models of recovery ([Lang et al., *Archives of Physical Medicine and Rehabilitation*, 2009; full text](#)). Therapist time is the limit, not the patient's day.

Devices that move the hand exist, but they are costly, clinic-bound or closed:

- Clinical robotic gloves such as Gloreha Sinfonia (now BTL R-TOUCH PRO) offer passive and active-assisted finger training, but they are tethered clinic systems ([Exoskeleton Report](#)).
- The NEOFECT Smart Glove costs about \$2,000 and tracks active movement for games; it does not move a hand that cannot move itself ([Recovery After Stroke](#)).
- Research tendon-driven gloves show the approach works. The Columbia hand orthosis for stroke uses a single forearm-mounted gearmotor with about 100 N peak tendon force and a dorsal tendon network to extend four fingers ([Park et al., *arXiv 1802.06131*; ROAM Lab](#)). Exo-Glove Poly II uses one actuator with antagonistic tendons and a 104 g polymer glove, but its actuation unit weighs about 1.14 kg and sits on a desk or wheelchair ([Kang et al., *Soft Robotics*, 2019](#)).

No open, garage-buildable, wearable design exists that a clinic, a research group or a family can build, inspect and adapt for daily repetitive finger movement at home.

A caution sits at the center of this problem. Robotic passive mobilization has been studied mainly for short-term effects on spasticity and limb perfusion ([Gobbo et al., "Hand Passive Mobilization Performed with Robotic Assistance: Acute Effects on Upper Limb Perfusion and Spasticity in Stroke Survivors," *BioMed Research International*, 2017; PMC5637828](#)). Whether passive repetitions alone improve hand function is not established. FlexHand is a passive-motion device (decided by Amish, 2026-09-25, FXH-DDR-001, D3); active or intent-triggered modes are a later option, and it makes no therapeutic claim.

■ Users and context

User	Need	Context
Stroke survivor with moderate hand impairment and mild flexor tone (MAS 1 to 1+) (subacute or chronic)	Many gentle, repeatable finger movements a day without waiting for a therapist	Home, seated with the forearm resting on a table or pillow; 30 to 60 min sessions, one to three a day
Care partner	Put the device on and take it off quickly and safely; know what to do if something goes wrong	Home
Occupational or physical therapist	Set range-of-motion, speed and force limits; see how many cycles were done	Clinic visit, then remote review
Researcher or open hardware community	A documented, reproducible platform to study dose and control strategies	University labs, makerspaces

■ Constraints

- Garage-buildable prototype, with a value-engineering target of USD 500 in parts (a control target, not a limit), using off-the-shelf motors and modules and 3D-printed parts.
- Wearable during a seated session: the hand side must be light and the actuators must sit on the forearm, not the hand.
- Fits a range of adult hands without custom molding.
- Safe with users who may have reduced sensation, spasticity and limited ability to remove the device themselves.
- Research and educational use only. FlexHand is not a medical device, has not been cleared or approved by any regulator, and must not be used to diagnose or treat any person outside a supervised research setting.

Safety: The device applies force to joints that may be spastic, contracted or insensate, and it carries a lithium-ion battery. Force limits, a physical stop, a tool-free tendon release and supervised use are part of the problem definition, not extras.

■ Out of scope

- Thumb actuation (the thumb is held in a passive abducted position in this concept).
- Wrist actuation.
- Diagnosis, assessment scoring or any therapeutic claim.
- Cloud services.

■ Open questions

- Which clinical partner to co-design with first (a stroke unit occupational therapy team, a community rehabilitation service, or a university rehabilitation lab)? Proposed, awaiting Amish. Decided on 2026-09-25 by Amish (FXH-DDR-001): passive motion only for this design, with active assist recorded as a later option (D3). The pitch is unchanged.

Decided on 2026-09-25 by Amish (FXH-DDR-002, N1): the first severity band is mild flexor tone (MAS 1 to 1+), which the 30 N design load covers according to the published finger stiffness check in FXH-CAL-001. Moderate and severe tone need roughly 80 to 150 N per finger and are left for a later design.